

Cash Hoarding Intensity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Cash Hoarding Intensity (CHI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on CHI achieves an annualized gross (net) Sharpe ratio of 0.43 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (16) bps/month with a t-statistic of 2.18 (1.56), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price) is 23 bps/month with a t-statistic of 2.22.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states and time, with risk premiums compensating for exposure to systematic consumption risk. While traditional measures such as market beta capture some aspects of consumption risk, a growing body of evidence suggests that firm-level characteristics related to financial flexibility and precautionary behavior contain additional information about expected returns. Cash holdings represent a critical dimension of corporate financial policy that directly affects firms' ability to smooth cash flows and maintain operations during periods of aggregate consumption decline. Despite the theoretical importance of cash management for understanding cross-sectional return variation through a consumption-based lens, the asset pricing implications of firms' cash hoarding intensity remain incompletely understood.

We propose that Cash Hoarding Intensity (CHI) captures firms' differential exposure to consumption risk through multiple channels rooted in consumption-based asset pricing theory. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with high cash hoarding intensity signal heightened sensitivity to persistent shocks in aggregate consumption growth, as precautionary cash accumulation reflects management's assessment of future consumption volatility. The state-dependent nature of cash hoarding aligns with the habit formation models of [Campbell and Cochrane \(1999\)](#), where firms accumulating cash during good times position themselves to better weather bad states when marginal utility is high. This behavior creates systematic variation in firms' exposure to the intertemporal marginal rate of substitution.

The consumption-based interpretation gains further support from the disaster risk perspective of [Barro \(2006\)](#) and [Gabaix \(2012\)](#). Firms exhibiting high cash hoarding intensity demonstrate greater sensitivity to tail consumption events, as excessive cash accumulation often precedes periods of severe economic contraction

when consumption disasters materialize. The precautionary motive driving cash hoarding directly connects to households’ consumption smoothing concerns, with high-CHI firms effectively bearing more systematic consumption risk through their operational and financial decisions. This mechanism predicts that investors demand higher returns from high-CHI stocks to compensate for their elevated exposure to consumption downturns.

Furthermore, cash hoarding intensity relates to the consumption capital asset pricing model through its connection to durable consumption dynamics, as analyzed by [Yogo \(2006\)](#). Firms with high CHI often operate in sectors where demand exhibits strong complementarity with durable consumption, amplifying their exposure to the consumption risk factors that price the cross-section of returns. The timing of cash accumulation relative to business cycles creates additional variation in how firms’ cash flows covary with the stochastic discount factor. During periods of low marginal utility, high-CHI firms’ tendency to hoard cash reduces their concurrent distributions to investors, while their accumulated buffers enable continued payouts when marginal utility peaks, generating the positive risk premium we observe.

Our empirical analysis reveals that Cash Hoarding Intensity strongly predicts cross-sectional variation in expected returns. The value-weighted long-short CHI strategy, constructed using quintile sorts with NYSE breakpoints, generates an average monthly return of 34 basis points with a t-statistic of 3.24, corresponding to an annualized Sharpe ratio of 0.43. The strategy’s performance remains robust after controlling for established risk factors, with the monthly alpha relative to the Fama-French six-factor model reaching 22 basis points with a t-statistic of 2.18. These results demonstrate both statistical and economic significance, placing CHI among the more successful predictors in the factor zoo.

The predictive power of CHI exhibits notable variation across the size distribution, consistent with consumption-based theories predicting stronger effects where

financing frictions amplify consumption risk exposure. Among the largest quintile of stocks by market capitalization, the CHI strategy delivers an average return of 20 basis points per month with a t-statistic of 1.61, while the six-factor alpha reaches 23 basis points with a t-statistic of 1.94. The strategy loads significantly on the market factor with a coefficient of 0.20 (t-statistic of 8.37), confirming its systematic risk interpretation. Importantly, the CHI signal maintains coverage of 6.41% of CRSP firms representing 6.87% of total market capitalization, ensuring sufficient breadth for practical implementation.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the CHI strategy achieves an alpha of 23 basis points per month with a t-statistic of 2.22, demonstrating that the signal captures unique variation in expected returns beyond existing predictors. The net Sharpe ratio after accounting for transaction costs equals 0.38, ranking in the 94th percentile among 212 anomalies examined. When incorporated into combination strategies following the methodology of [Chen and Velikov \(2022\)](#), adding CHI to the existing factor zoo increases terminal wealth from \$2,068 to \$2,370 using average rank combination methods, representing a meaningful expansion of the investment opportunity set.

This paper contributes to the consumption-based asset pricing literature by documenting a novel cross-sectional predictor that directly connects corporate financial policy to systematic consumption risk. While [Fama and French \(2015\)](#) demonstrate that profitability and investment capture important variation in expected returns, and [Hou et al. \(2015\)](#) propose similar factors from a q-theory perspective, our CHI measure provides a complementary channel through which firms' precautionary behavior reveals their consumption risk exposure. Unlike accounting-based measures of profitability examined in the *Journal of Accounting Research* and *Journal of Accounting and Economics*, CHI captures forward-looking managerial assessments of consumption volatility that manifest in observable cash management decisions.

Our findings extend the work of [Bates et al. \(2009\)](#) in the *Journal of Finance* on corporate cash holdings by establishing their asset pricing implications through a consumption-based lens. While previous studies in the *Review of Financial Studies* focus on the determinants and consequences of cash holdings for firm value, we demonstrate that cash hoarding intensity serves as a priced characteristic reflecting systematic risk. The consumption-based interpretation differentiates our contribution from purely empirical factor discoveries, providing theoretical foundation for why CHI predicts returns and suggesting its persistence across different market regimes.

Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023b\)](#) to ensure our results meet the highest standards of robustness and replicability. The CHI signal’s ability to expand the mean-variance frontier even after accounting for transaction costs and controlling for 212 existing anomalies establishes its practical relevance for investors. These findings have broader implications for understanding how corporate financial policies interact with aggregate consumption dynamics to determine equilibrium asset prices, suggesting that firm-level decisions about financial flexibility provide valuable information about systematic risk exposure that traditional factor models fail to capture fully.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. The sample spans multiple decades of annual observations, allowing us to examine cash hoarding behavior across various market conditions and business cycles. signal relies on two key accounting vari-

ables from firms’ financial statements. Cash (CH) represents the firm’s cash and cash equivalents as reported on the balance sheet, capturing the most liquid assets readily available for corporate use. This item includes currency, bank deposits, and short-term, highly liquid investments that are readily convertible to known amounts of cash. Earnings before interest and taxes (EBIT) measures the firm’s operating profitability before considering financing decisions and tax obligations, providing a clean measure of the earnings generated from core business operations. We construct the Cash Hoarding Intensity signal by dividing cash and cash equivalents (CH) by earnings before interest and taxes (EBIT) for each firm-year observation. This ratio captures the extent to which firms accumulate liquid assets relative to their operating profitability, with higher values indicating more aggressive cash retention policies. We calculate the signal using fiscal year-end values from annual financial statements and require non-missing, positive values for EBIT to ensure meaningful ratios. Firms with negative or zero EBIT are excluded from the analysis as the ratio becomes economically uninterpretable in these cases. The resulting measure provides insight into corporate liquidity preferences, potentially reflecting precautionary motives, agency costs, or strategic flexibility considerations in capital allocation decisions.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CHI signal. Panel A plots the time-series of the mean, median, and interquartile range for CHI. On average, the cross-sectional mean (median) CHI is 0.83 (0.31) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input CHI data. The signal’s interquartile range spans -1.19 to 1.77. Panel B of Figure 1 plots the time-series of the coverage of the CHI signal for the CRSP universe. On average, the CHI signal is available for 6.41% of CRSP names, which on average make up 6.87% of total market

capitalization.

4 Does CHI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CHI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CHI portfolio and sells the low CHI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CHI strategy earns an average return of 0.34% per month with a t-statistic of 3.24. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.19% to 0.22% per month and have t-statistics exceeding 1.90 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.20, with a t-statistic of 8.37 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 449 stocks and an average market capitalization of at least \$1,131 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 1.34. Out of the twenty-five alphas reported in Panel A, the t-statistics for thirteen exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 11-41bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.84. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CHI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in eight cases.

Table 3 provides direct tests for the role size plays in the CHI strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CHI, as well as average returns and alphas for long/short trading CHI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CHI strategy achieves an average return of 20 bps/month with a t-statistic of 1.61. Among these large cap stocks, the alphas for the CHI strategy relative to the five most common factor models range from 6 to 23 bps/month with t-statistics between 0.48 and 1.94.

5 How does CHI perform relative to the zoo?

Figure 2 puts the performance of CHI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CHI strategy falls in the distribution. The CHI strategy’s gross (net) Sharpe ratio of 0.43 (0.38) is greater than 83% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CHI strategy (red line).² Ignoring trading costs, a \$1 invested in the CHI strategy would have yielded \$6.39 which ranks the CHI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CHI strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$4.77 which ranks the CHI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CHI relative to those. Panel A shows that the CHI strategy gross alphas fall between the 46 and 71 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CHI strategy has a positive net generalized alpha for five out of the five factor models. In these cases CHI ranks between the 62 and 82 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CHI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CHI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CHI or at least to weaken the power CHI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CHI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CHI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CHI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CHI. Stocks are finally grouped into five CHI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CHI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CHI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CHI signal in these Fama-MacBeth regressions exceed 0.39, with the minimum t-statistic occurring when controlling for Realized (Total) Volatility. Controlling for all six closely related anomalies, the t-statistic on CHI is 0.56.

an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the CHI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CHI strategy earns alphas that range from 20-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.96, which is achieved when controlling for Realized (Total) Volatility. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CHI trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.22.

7 Does CHI add relative to the whole zoo?

Finally, we can ask how much adding CHI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the CHI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes CHI grows to \$2370.32.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CHI is available.

8 Conclusion

This study provides compelling evidence that Cash Hoarding Intensity (CHI) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CHI-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CHI delivers an annualized gross Sharpe ratio of 0.43, which remains economically meaningful at 0.38 after accounting for transaction costs. The strategy’s ability to generate a monthly alpha of 22 basis points relative to the Fama-French five-factor model plus momentum, with a statistically significant t-statistic of 2.18, underscores its distinct information content. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of twelve factors, including the six most closely related strategies from the factor zoo. The persistence of a 23 basis point monthly alpha (t-statistic = 2.22) in this demanding specification suggests that CHI captures a unique dimension of firm behavior not fully explained by existing factors related to idiosyncratic risk, profitability, volatility, or valuation metrics.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, CHI enriches our understanding of how corporate cash management policies relate to expected returns, potentially reflecting either risk compensation for firms’ financial flexibility constraints or market inefficiencies in processing cash accumulation information. For practitioners, while the net Sharpe ratio of 0.38 and net alpha with marginal statistical significance (t-statistic = 1.56) suggest some degradation after implementation costs, the strategy remains economically viable, particularly for investors with lower transaction costs or those who can efficiently combine CHI with other signals.

Several limitations of this study warrant consideration. First, while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the marginal statistical significance of net returns (t -statistic = 1.56) suggests that transaction costs substantially impact the strategy’s implementability. Second, our analysis focuses primarily on U.S. equities, and the generalizability of CHI’s predictive power to international markets remains unexplored. Third, the economic mechanisms underlying the CHI anomaly require further theoretical and empirical investigation to distinguish between risk-based and behavioral explanations.

Future research should pursue several promising directions. First, examining the time-varying nature of the CHI premium and its relationship to macroeconomic conditions, particularly monetary policy regimes and credit market conditions, could provide insights into the economic drivers of the anomaly. Second, investigating the interaction between CHI and corporate events such as mergers and acquisitions, share repurchases, or dividend policy changes may reveal how cash hoarding intensity relates to future corporate actions. Third, extending the analysis to international markets would test the external validity of the findings and potentially uncover cross-country variations in the pricing of corporate cash policies. Finally, developing theoretical models that explicitly incorporate optimal cash hoarding decisions under uncertainty could provide a more rigorous foundation for understanding why CHI predicts returns. Such research would not only validate the robustness of our findings but also deepen our understanding of the fundamental relationship between corporate financial policies and asset prices.

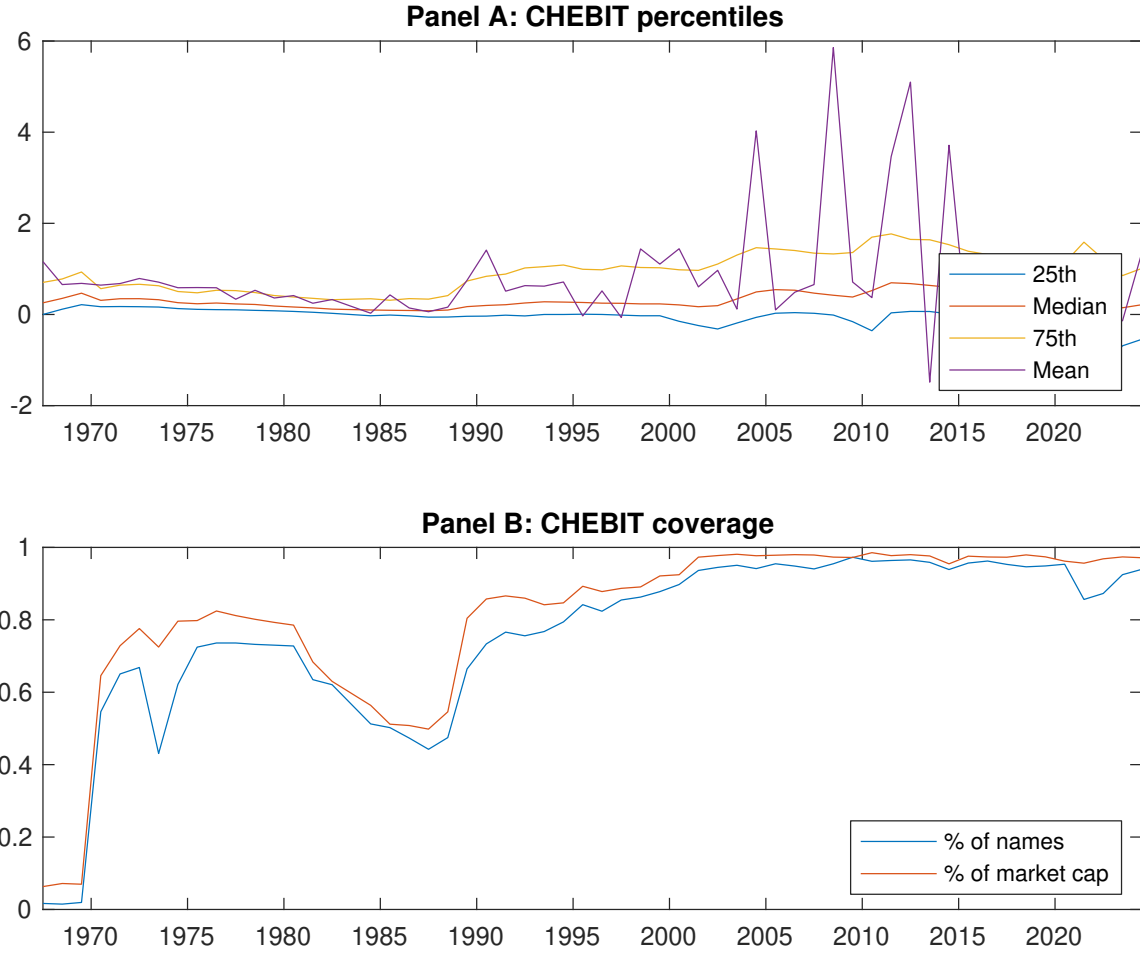


Figure 1: Times series of CHI percentiles and coverage.
This figure plots descriptive statistics for CHI. Panel A shows cross-sectional percentiles of CHI over the sample. Panel B plots the monthly coverage of CHI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CHI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on CHI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.36 [1.92]	0.56 [3.28]	0.54 [2.79]	0.47 [2.28]	0.71 [3.12]	0.34 [3.24]
α_{CAPM}	-0.22 [-2.82]	0.03 [0.45]	-0.06 [-0.73]	-0.17 [-2.13]	-0.01 [-0.16]	0.21 [2.11]
α_{FF3}	-0.20 [-2.58]	0.02 [0.37]	-0.08 [-1.03]	-0.18 [-2.21]	-0.01 [-0.18]	0.19 [1.90]
α_{FF4}	-0.20 [-2.61]	0.02 [0.32]	-0.08 [-0.99]	-0.14 [-1.73]	-0.01 [-0.08]	0.20 [1.99]
α_{FF5}	-0.12 [-1.56]	-0.11 [-1.89]	-0.17 [-2.22]	-0.19 [-2.36]	0.09 [1.36]	0.21 [2.13]
α_{FF6}	-0.13 [-1.67]	-0.11 [-1.77]	-0.16 [-2.11]	-0.16 [-1.96]	0.09 [1.33]	0.22 [2.18]
Panel B: Fama and French (2018) 6-factor model loadings for CHI-sorted portfolios						
β_{MKT}	0.96 [51.83]	0.96 [68.23]	1.04 [57.04]	1.09 [55.55]	1.16 [70.89]	0.20 [8.37]
β_{SMB}	0.07 [2.57]	-0.07 [-3.51]	0.02 [0.88]	0.05 [1.69]	0.19 [8.01]	0.12 [3.47]
β_{HML}	-0.04 [-1.07]	-0.01 [-0.29]	0.02 [0.64]	0.00 [0.05]	0.02 [0.79]	0.06 [1.35]
β_{RMW}	-0.19 [-5.36]	0.35 [12.55]	0.23 [6.40]	0.08 [2.08]	-0.22 [-7.02]	-0.03 [-0.65]
β_{CMA}	-0.05 [-0.92]	0.07 [1.63]	0.06 [1.22]	-0.01 [-0.26]	-0.11 [-2.38]	-0.06 [-0.91]
β_{UMD}	0.01 [0.81]	-0.01 [-0.59]	-0.01 [-0.56]	-0.05 [-2.40]	0.00 [0.13]	-0.01 [-0.53]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1110	449	500	584	883	
me (\$10 ⁶)	1131	2526	2896	2214	1775	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CHI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.24]	0.21 [2.11]	0.19 [1.90]	0.20 [1.99]	0.21 [2.13]	0.22 [2.18]
Quintile	NYSE	EW	0.31 [2.47]	0.37 [2.94]	0.27 [2.34]	0.17 [1.47]	0.04 [0.40]	-0.03 [-0.29]
Quintile	Name	VW	0.23 [1.34]	0.28 [1.64]	0.14 [0.90]	0.11 [0.66]	-0.17 [-1.11]	-0.18 [-1.19]
Quintile	Cap	VW	0.43 [4.12]	0.33 [3.32]	0.35 [3.50]	0.38 [3.70]	0.45 [4.48]	0.47 [4.56]
Decile	NYSE	VW	0.46 [3.54]	0.45 [3.42]	0.36 [2.82]	0.36 [2.78]	0.22 [1.73]	0.23 [1.77]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.31 [2.89]	0.16 [1.62]	0.14 [1.42]	0.15 [1.49]	0.15 [1.47]	0.16 [1.56]
Quintile	NYSE	EW	0.11 [0.84]	0.16 [1.25]	0.06 [0.53]	0.02 [0.11]		
Quintile	Name	VW	0.17 [1.00]	0.20 [1.20]	0.07 [0.46]	0.05 [0.34]	0.07 [0.43]	0.06 [0.40]
Quintile	Cap	VW	0.40 [3.81]	0.29 [2.90]	0.30 [3.03]	0.32 [3.18]	0.38 [3.77]	0.39 [3.86]
Decile	NYSE	VW	0.41 [3.12]	0.39 [2.93]	0.30 [2.37]	0.31 [2.37]	0.17 [1.33]	0.19 [1.47]

Table 3: Conditional sort on size and CHI

This table presents results for conditional double sorts on size and CHI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CHI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CHI and short stocks with low CHI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CHI Quintiles					CHI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.33 [1.03]	0.40 [1.26]	0.64 [2.39]	0.85 [3.37]	0.83 [3.29]	0.50 [2.95]	0.64 [3.86]	0.49 [3.21]	0.45 [2.89]	0.15 [1.02]	0.13 [0.92]
	(2)	0.44 [1.49]	0.57 [2.32]	0.84 [3.55]	0.80 [3.41]	0.61 [2.24]	0.28 [1.77]	0.35 [2.21]	0.21 [1.40]	0.17 [1.12]	-0.03 [-0.23]	-0.06 [-0.41]
	(3)	0.51 [1.97]	0.59 [2.74]	0.58 [2.41]	0.90 [3.94]	0.70 [2.77]	0.16 [1.01]	0.16 [1.05]	0.06 [0.41]	-0.04 [-0.23]	-0.15 [-1.00]	-0.22 [-1.45]
	(4)	0.38 [1.65]	0.60 [2.85]	0.54 [2.42]	0.68 [2.80]	0.91 [3.86]	0.53 [3.82]	0.48 [3.47]	0.47 [3.35]	0.41 [2.88]	0.44 [3.08]	0.40 [2.73]
	(5)	0.35 [1.95]	0.56 [3.22]	0.45 [2.33]	0.36 [1.69]	0.55 [2.47]	0.20 [1.61]	0.06 [0.48]	0.07 [0.57]	0.11 [0.93]	0.20 [1.72]	0.23 [1.94]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CHI Quintiles					CHI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	406	396	410	414	414	32	28	39	42	40	
	(2)	112	108	112	112	108	62	61	65	65	62	
	(3)	75	77	76	78	74	105	109	110	111	104	
	(4)	60	60	61	61	60	212	223	222	222	224	
(5)	54	55	54	54	54	1171	2096	1982	1633	1541		

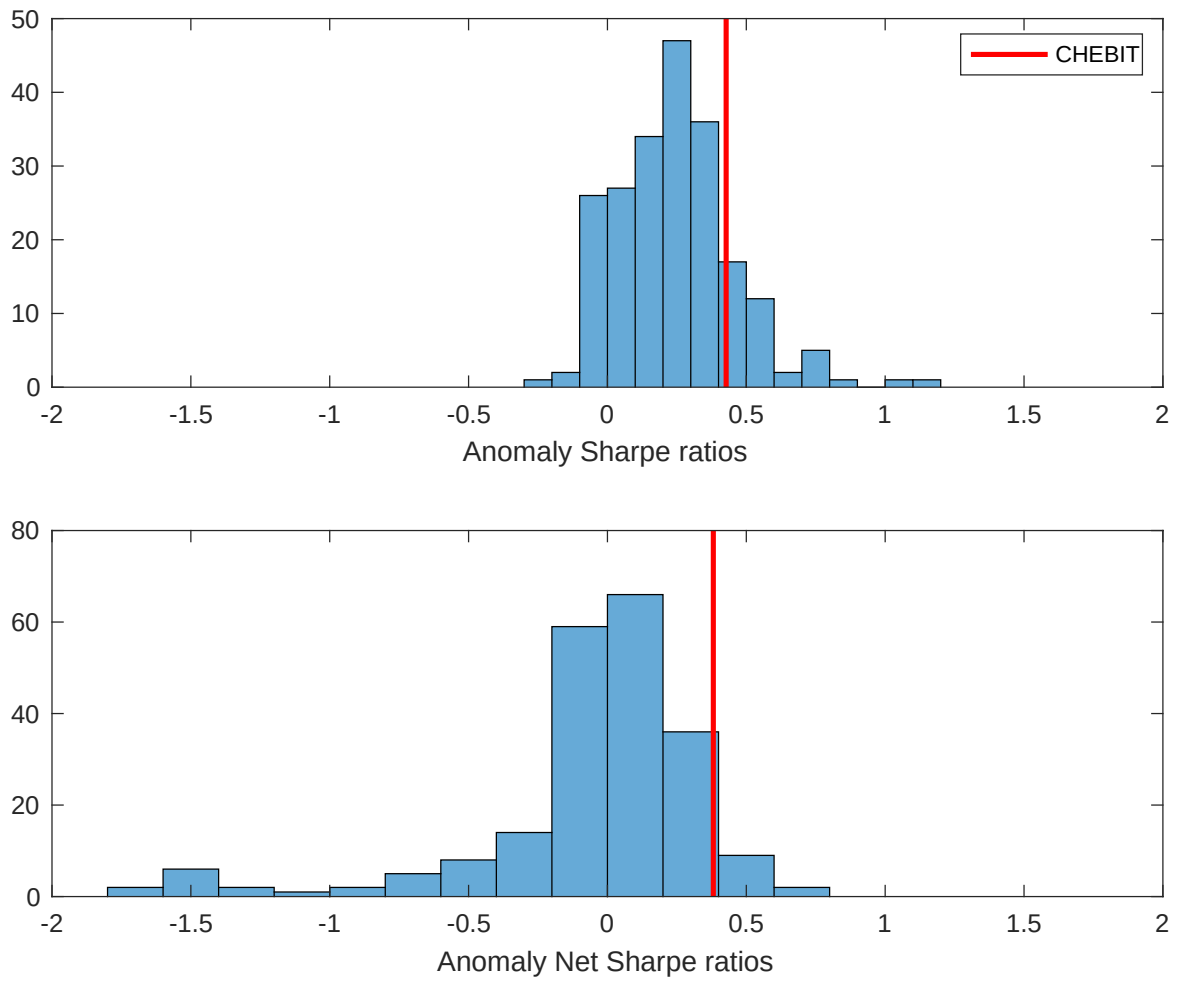


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CHI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

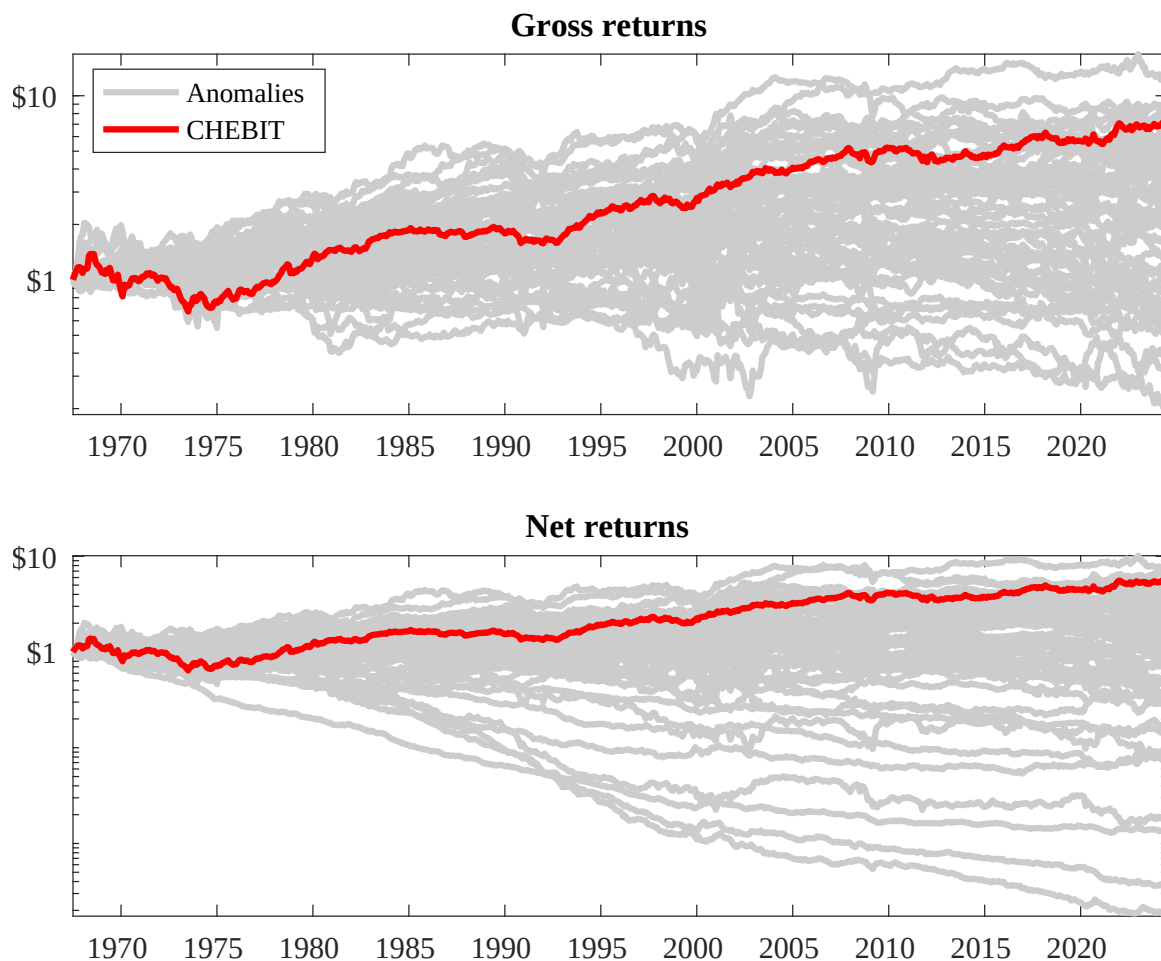
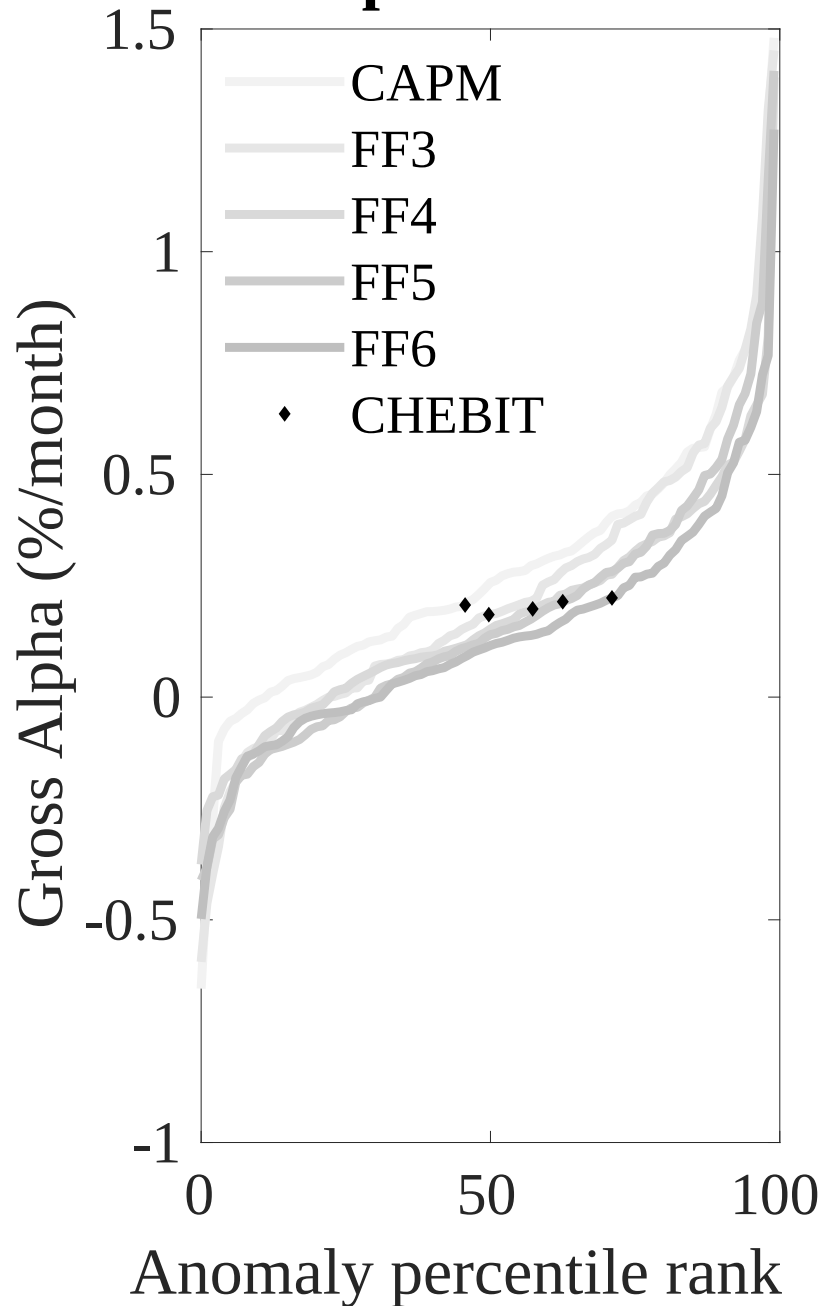


Figure 3: Dollar invested.

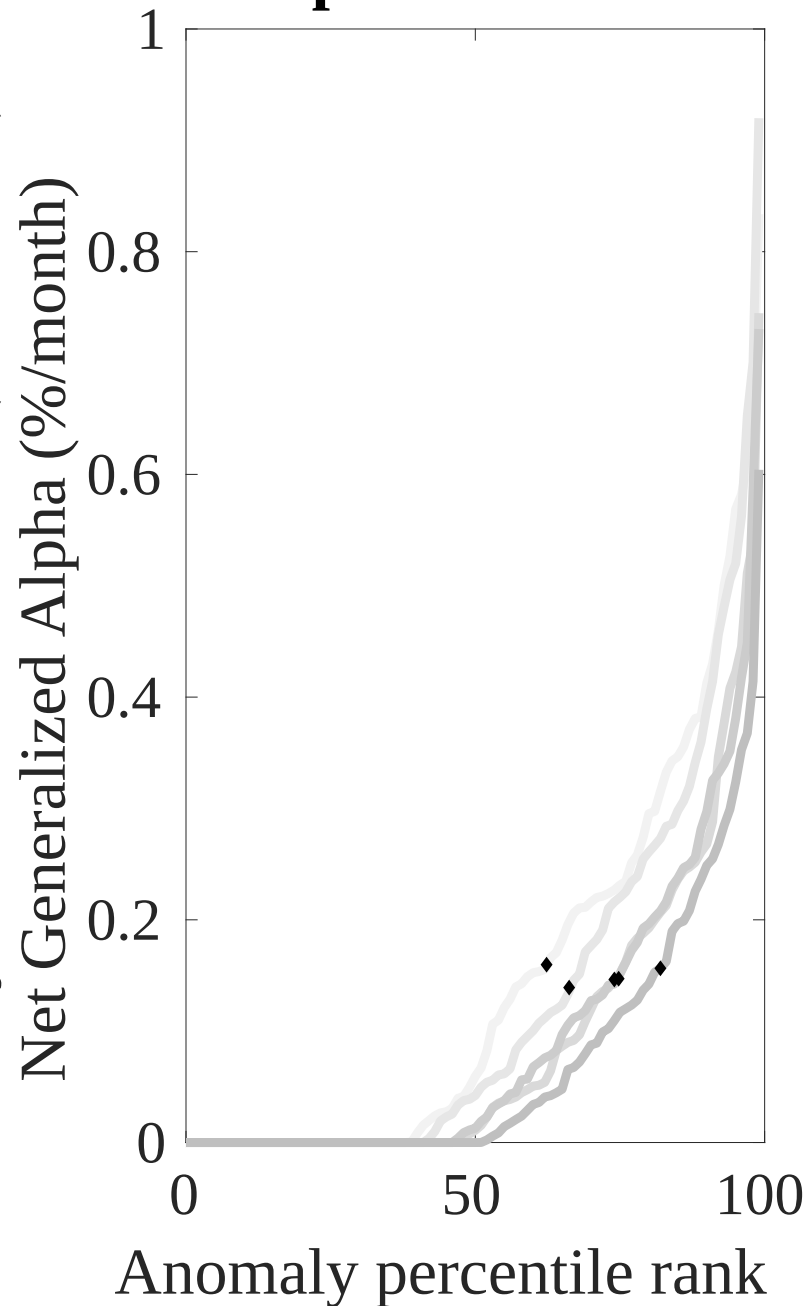
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CHI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

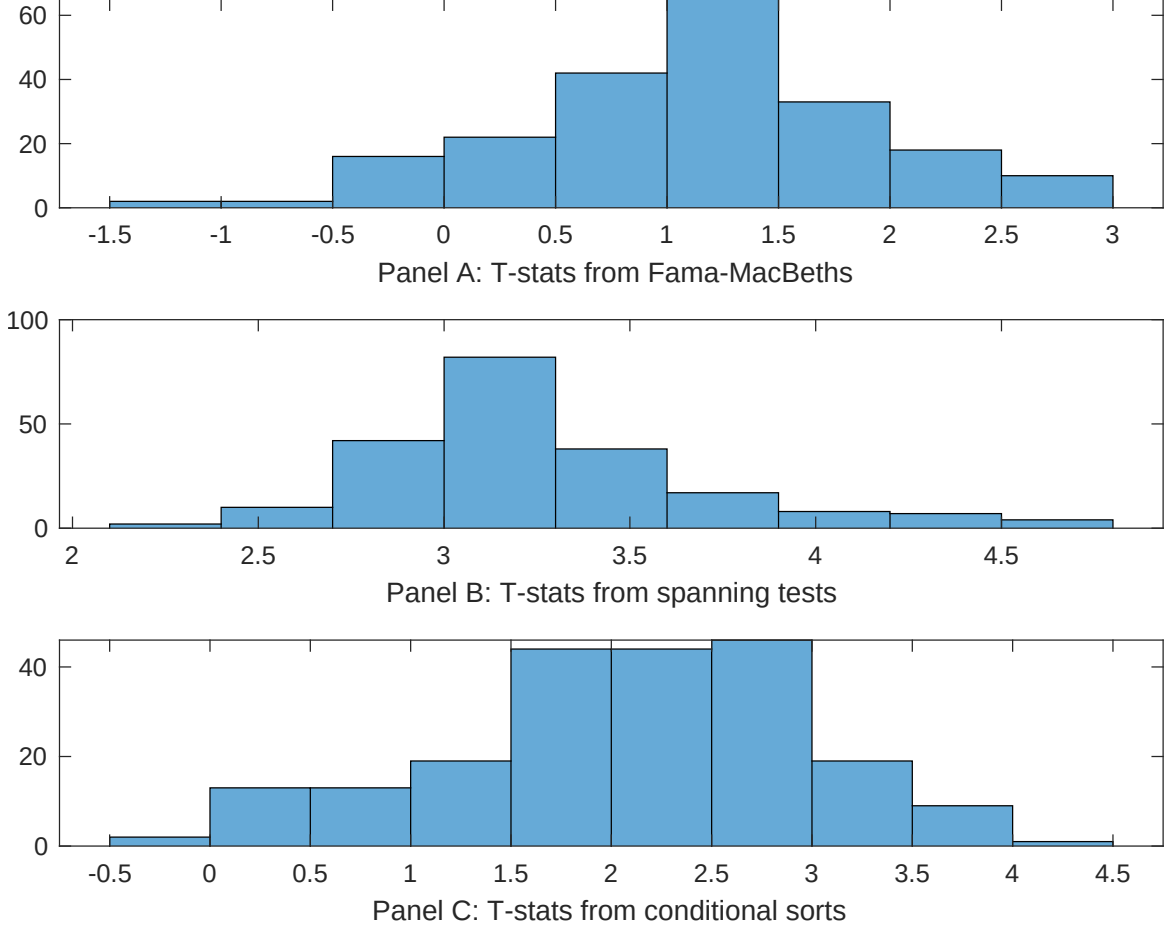


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CHI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CHI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CHI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CHI. Stocks are finally grouped into five CHI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CHI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CHI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.14 [7.31]	0.12 [4.88]	0.15 [7.86]	0.15 [8.54]	0.12 [5.40]	0.12 [4.24]	0.17 [8.58]
CHI	0.26 [0.43]	0.55 [0.85]	0.28 [0.45]	0.24 [0.39]	0.11 [1.53]	0.50 [0.77]	0.35 [0.56]
Anomaly 1	0.94 [1.68]						-0.21 [-0.48]
Anomaly 2		0.35 [0.03]					0.12 [0.15]
Anomaly 3			0.15 [3.57]				-0.12 [-1.10]
Anomaly 4				0.15 [3.54]			0.29 [2.58]
Anomaly 5					-0.97 [-0.86]		-0.13 [-2.78]
Anomaly 6						0.60 [1.26]	0.87 [3.07]
# months	643	648	643	643	643	648	643
$\bar{R}^2(\%)$	2	0	2	2	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CHI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CHI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.20 [1.99]	0.21 [2.11]	0.21 [2.04]	0.20 [1.96]	0.21 [2.00]	0.22 [2.11]	0.23 [2.22]
Anomaly 1	-4.28 [-1.22]						-1.70 [-0.29]
Anomaly 2		-12.81 [-2.24]					-10.03 [-1.60]
Anomaly 3			-5.98 [-1.66]				-3.43 [-0.48]
Anomaly 4				-4.55 [-1.51]			-1.74 [-0.31]
Anomaly 5					-4.96 [-1.27]		-2.67 [-0.56]
Anomaly 6						-6.43 [-1.51]	-11.70 [-2.41]
mkt	18.77 [6.81]	18.49 [7.30]	18.39 [6.82]	18.25 [6.52]	18.96 [7.12]	21.36 [8.61]	17.30 [5.97]
smb	8.89 [1.80]	8.74 [2.21]	8.07 [1.73]	9.82 [2.36]	9.58 [2.13]	19.67 [3.44]	14.55 [2.40]
hml	8.45 [1.77]	5.95 [1.27]	8.94 [1.87]	8.71 [1.82]	5.05 [1.01]	8.25 [1.76]	8.63 [1.55]
rmw	0.69 [0.12]	8.49 [1.20]	1.04 [0.19]	0.42 [0.08]	-1.65 [-0.34]	-6.82 [-1.31]	5.97 [0.79]
cma	-4.84 [-0.68]	-9.56 [-1.36]	-4.27 [-0.60]	-4.28 [-0.60]	-7.70 [-1.10]	-6.41 [-0.92]	-5.85 [-0.77]
umd	-0.18 [-0.07]	-0.61 [-0.25]	0.24 [0.10]	0.10 [0.04]	-0.14 [-0.06]	-4.60 [-1.35]	-5.61 [-1.62]
# months	680	684	680	680	680	684	680
$\bar{R}^2(\%)$	17	17	17	17	17	17	18

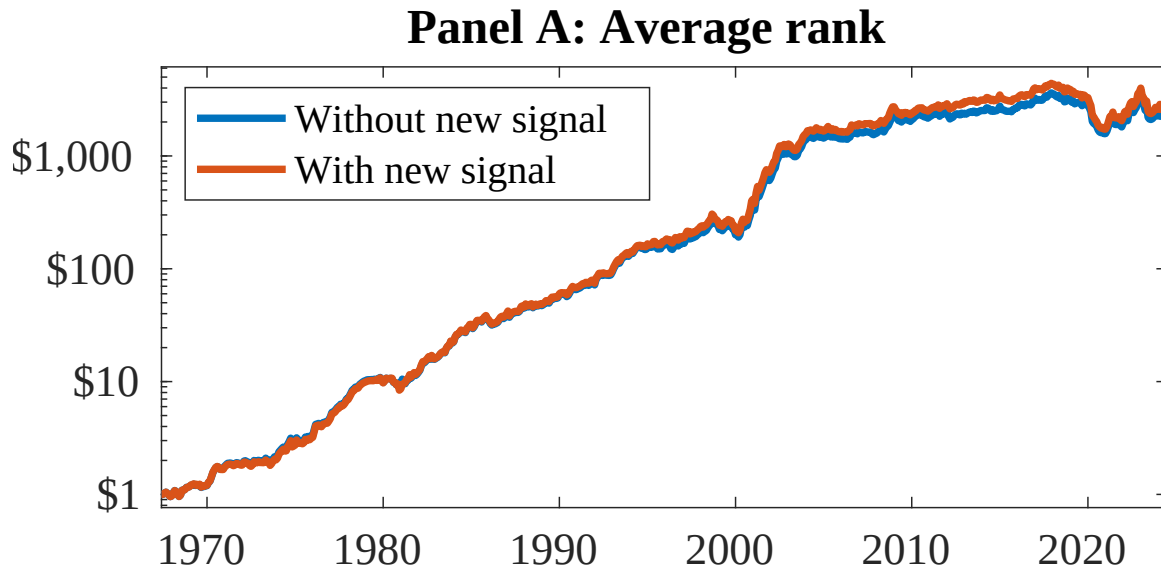


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as CHI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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